

Intelligent Oscillometric System for Automatic Detection of Peripheral Arterial Disease

Nosratallah Forghani¹, Keivan Maghooli, Nader Jafarnia Dabanloo, Ali Vasheghani Farahani, and Mohamad Forouzanfar², *Senior Member, IEEE*

Abstract—Peripheral arterial disease (PAD) is a progressing arterial disorder that is associated with significant morbidity and mortality. The conventional PAD detection methods are invasive, cumbersome, or require expensive equipment and highly trained technicians. Here, we propose a new automated, noninvasive, and easy-to-use method for the detection of PAD based on characterizing the arterial system by applying an external varying pressure using a cuff. The superposition of the internal arterial pressure and the externally applied pressure were measured and mathematically modeled as a function of cuff pressure. A feature-based learning algorithm was then designed to identify PAD patterns by analyzing the parameters of the derived mathematical models. Genetic algorithm and principal component analysis were employed to select the best predictive features distinguishing PAD patterns from normal. A RUSBoost ensemble model using neural network as the base learner was designed to diagnose PAD from genetic algorithm selected features. The proposed method was validated on data collected from 14 PAD patients and 19 healthy individuals. It achieved a high accuracy, sensitivity, and specificity of 91.4%, 90.0%, and 92.1%, respectively, in detecting PAD. The effect of age, a confounding factor that may have impacted our analyzes, was not considered in this study. The proposed method shows promise toward noninvasive and accurate detection of PAD and can be integrated into routine oscillometric blood pressure measurements.

Index Terms—Peripheral artery disease (PAD), Oscillometry, Physiological Monitoring, Neural network (NN), Genetic Algorithm (GA), Feature extraction, Classification, Diagnosis.

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Nosratallah Forghani, Keivan Maghooli, and Nader Jafarnia Dabanloo are with Department of Biomedical Engineering, Science and Research Branch, Islamic Azad University, Tehran 14778-93855, Iran (e-mail: n.forghani@srbiau.ac.ir; k_maghool@srbiau.ac.ir; n_jafarnia@srbiau.ac.ir).

Ali Vasheghani Farahani is with the Cardiac Primary Prevention Research Center, Cardiovascular Disease Research Institute, Tehran University of Medical Sciences, Tehran 14174-66191, Iran (e-mail: avasheghani@tums.ac.ir).

Mohamad Forouzanfar is with the Department of Systems Engineering, École de technologie supérieure (ÉTS), Université du Québec, Montreal, QC H3C 1K3, Canada, and also with the Department of Biomedical Engineering, K.N. Toosi University of Technology, Tehran 19967-15433, Iran (e-mail: mohamad.forouzanfar@etsmtl.ca, forouzanfar@kntu.ac.ir).

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I. INTRODUCTION

PERIPHERAL arterial disease (PAD) is a serious circulatory problem caused by building fatty plaques in the walls of the peripheral arteries. The accumulation of fatty plaques thickens and hardens the arteries, reduces arteries elasticity, and affects normal blood flow to organs and tissues [1], [2]. PAD is a manifestation of widespread atherosclerosis that is associated with an increased risk of myocardial infarction, coronary artery disease, heart attack, and stroke [3]. The prevalence of PAD increases with age, especially from the fourth or fifth decade of life [3] and significantly affects the quality of life by developing limb ischemia and limiting physical activities. About 20% of people over 60 years old in the world are affected by PAD [4]. In the United States, PAD affects 8 to 12 million people out of which 30% remain undiagnosed [5]. Early detection of PAD can help in reducing the severity of the disease and the need for possible amputations.

The most accurate techniques for the detection of PAD include arteriography (angiography) and intravascular ultrasound imaging techniques [6], [7]. Given the invasive nature of these techniques and their inherent risks, they are not suitable for early detection and routine screening of PAD. The most common noninvasive techniques for the detection of PAD include vascular ultrasound [8], blood pressure measurement [9], and plethysmography [10]–[12].

Vascular ultrasound is based on the measurement of speed of blood flow by transmitting ultrasound waves and recording the reflected waves off the blood cells moving within the vessels [13]. Faster speed of blood flow indicates narrowing or blockage of the vessels. Vascular ultrasound is highly operator dependent, cumbersome, and time-consuming, and may not work for patients with heavily calcified arteries and obesity [13].

Blood pressure measurement technique is based on comparing the systolic pressure measured at the ankle with systolic pressure measured at the arm by calculating an ankle-brachial index (ABI) [9]. Currently, the resting ABI is the initial diagnostic test recommended by the 2016 American Heart Association/American College of Cardiology (AHA/ACC) guidelines [14]. A low ABI value indicates narrowing or blockage of the peripheral arteries. Systolic pressure is conventionally estimated using auscultation method where an inflatable cuff is placed around the individual's arm or ankle and is deflated from a presystolic pressure to a sub diastolic pressure while an expert is listening to the Korotkoff sounds within the arteries

using a stethoscope. Blood pressure may also be estimated automatically using a similar technique to auscultation called oscillometry [15]. The auscultation-based ABI technique is time consuming and requires expertise while the automated oscillometric-based ABI technique is not reliable as the accuracy of oscillometric blood pressure monitors significantly decrease in patients with PAD.

Plethysmography based techniques measure blood volume changes in a tissue over each cardiac cycle. The most common plethysmography techniques are the impedance plethysmography (IPG) [10] and photo plethysmography (PPG) [3], [11], [12], [16]. The variations in the morphological features of the measured blood volume over each cardiac cycle from their normal patterns are analyzed to detect PAD. Recently, deep learning has been employed to detect peripheral arterial disease from pulse waveform [17], [18]. In [19], [20], a transfer function was built between peripheral and central arterial waveform for the detection of arterial disease. However, the morphology of the peripheral blood volume may change due to several environmental and physiological factors other than arterial disease and may not provide a reliable measure for the detection of PAD [3], [21].

The application of current PAD detection methods is limited in the sense that they are invasive [6], [7], cumbersome and time-consuming [10], [22], require expensive equipment and highly trained technicians [13], or they are not always reliable [10], [11], [15]. Given that up to 50% percent of patients with PAD are symptomless [23], new automated, affordable, and easy-to-use diagnostic methods are needed for the early detection of PAD.

In this paper, we present a new intelligent noninvasive approach for the detection of PAD that can be easily performed with an oscillometric cuff without the need for any special expertise. Our proposed technique is based applying an external time varying pressure to the ankles of individuals using an inflatable cuff and characterizing the arterial system response at different external pressures. The superimposed arterial pressure pulses over the applied cuff pressure are measured using a pressure transducer and are mathematically modeled as a function of cuff pressure. Feature extraction algorithms based on genetic algorithm (GA) and principal component analysis are used to extract the best predictive features out of the model parameters. An ensemble algorithm based on neural networks (NN) is then designed to detect PAD patterns from the extracted features. The performance of the proposed method is validated on the detection of PAD in a pilot investigation on data collected from 14 PAD patients and 19 healthy individuals.

II. METHODOLOGY

A. Experimental Setup

Nineteen healthy adults and fourteen patients with PAD were recruited from Sina Hospital, Tehran, Iran. PAD disease was diagnosed using the gold standard angiography technique by a vascular specialist. Among patients with PAD disease, three were diagnosed with PAD only in right leg, two only in left

TABLE I
CHARACTERISTICS OF PARTICIPANTS AND RESTING-STATE
CARDIOVASCULAR MEASURES

Characteristics	Healthy	PAD
Total Number	19 (9 female)	14 (6 female)
Age	45.2± 13.6	58.3± 17.1
SBP	133.4± 26.0	143.1± 25.5
DBP	88.78± 17.7	90.1± 15.6
HR	74.5± 10.5	87.3± 16.3
ABI	1.24± 0.14	0.89± 0.41

leg, and nine in both legs. All recordings were performed in the Cardiovascular Clinic at Sina Hospital. The study was reviewed and approved by the Research Ethics Committee of Azad University- Science and Research Branch, and all participants provided written informed consent. Participant's characteristics are provided in Table I.

PAD was initially diagnosed on individuals who had leg discomfort with exercise by performing different tests, including physical examinations (weak or absent pulse below a narrowed area of the artery, evidence of poor wound healing in the area where the blood flow is restricted, and decreased blood pressure in the affected limb), ABI test, and blood test (to measure cholesterol and triglycerides and to check for diabetes). These subjects were then referred by the specialist to perform an angiography test for the final diagnosis. All the patients with PAD in our study had performed an angiography test and the presence of PAD in at least one of their lower limbs has been confirmed by the specialist.

An integrated set of equipment was used to characterize the arterial system behavior in different external cuff pressures. Our measurement system consisted of three main components: 1) automatic cuff-based blood pressure measurement system to control the inflation and deflation of an oscillometric cuff (Omron 7321 M6 Comfort Digital Blood Pressure Monitor, Omron Co., Kyoto, Japan); 2) analog to digital converter module (Biopac Universal Interface Module) (16 Bits resolution); and 3) a computer system equipped with Biopac AcqKnowledge data acquisition and analysis software (sampling rate set to 500 Hz).

Participants were seated comfortably on a bed. The inflatable cuff was placed around the limb under study. The cuff was inflated to a high pressure (above systolic pressure), gradually deflated at a rate of 2 mmHg/s to a low pressure (below diastolic pressure), and then completely deflated. The superposition of the externally applied cuff pressure and internal arterial pressure was measured using a transducer within the cuff. The sensor operational pressure was from -50 mmHg to +300 mmHg, its dynamic response was 100 Hz, and the combined effects of sensitivity, linearity, and hysteresis of the sensor were 1mmHg (nominal). Measurements were performed sequentially on the left arm, right arm, left ankle, and right ankle, respectively. The recordings were performed twice on each subject with 10 mins delay in between. This led to a total of 264 waveforms. Along with each recording, the Omron estimated blood pressure was also recorded. All the measurements were performed under medical expert supervision.

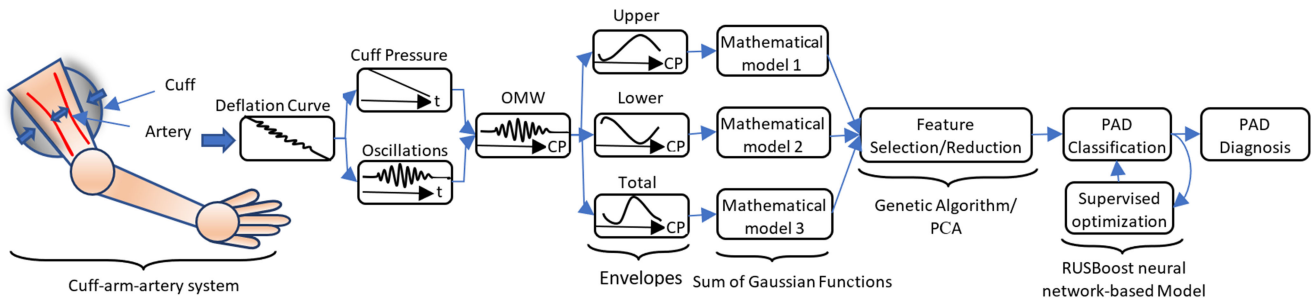


Fig. 1. Characterizing arterial system behavior at different external cuff pressures and detecting PAD.

B. Arterial System Characterization

Human arteries affected by PAD lose their elastic properties and become stiffened over time. To characterize the arterial elastic properties, an external time varying pressure was applied to the artery under study and its behavior at different external pressures was evaluated. The concept of applying an external time varying pressure to characterize the arterial system behavior, was originally designed to estimate mean arterial pressure and later on extended to the estimation of systolic and diastolic pressures [24]. Through several studies it has been shown that the approach can characterize the arteries elastic properties including arterial collapse and distension [25]–[27]. Here, we investigated the applicability of oscillometric technique for the noninvasive and convenient detection of PAD.

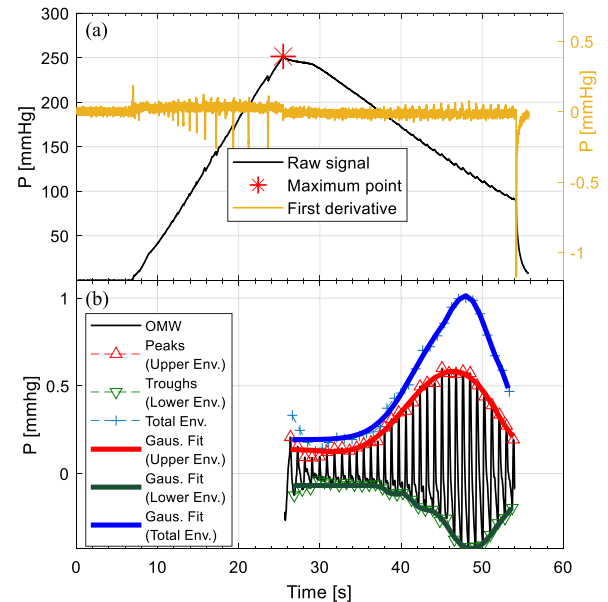
A block diagram representation of our proposed arterial system characterization approach for PAD detection is shown in Fig. 1 and is discussed below.

Oscillometric Waveform Measurement: Fig. 2 illustrates examples of the recorded cuff pressures for a healthy individual and a patient with PAD. All the processing steps are shown in the figure and described in the following.

The measured pressure within the cuff (Fig. 2(a)) was pre-processed to extract the deflation curve, the waveform that represents the gradual deflation of cuff pressure. The time point at which the maximum measured pressure occurred was selected as the onset of deflation process and the time point at which the derivative of the measured pressure exhibited a minimum after the deflation onset was selected as the deflation process offset. These two time points were then used to extract the deflation curve.

Deflation curve consists of a slow varying pattern representing the applied cuff pressure and an oscillatory pattern caused by the arterial lumen area variations at different transmural pressures [25], [26]. The slow varying cuff pressure was extracted from the recorded cuff deflation curve using a lowpass 4th-order Butterworth filter with a cutoff frequency of 0.5 Hz. The extracted waveform represents an estimate of the applied cuff pressure. The arterial lumen area oscillations were extracted from the recorded deflation curve using a highpass 4th-order Butterworth filter with a cutoff frequency of 0.8 Hz. The extracted oscillations over the cuff deflation period are called the oscillometric waveform (OMW) (Fig. 2(b)). The amplitude of the oscillometric pulses increase to a peak and then decrease with further deflation and the maximum oscillation occurs when the artery is unloaded,

Healthy Individual - Male - 62 Years Old - Right Leg:



PAD Individual - Male - 64 Years Old - Right leg:

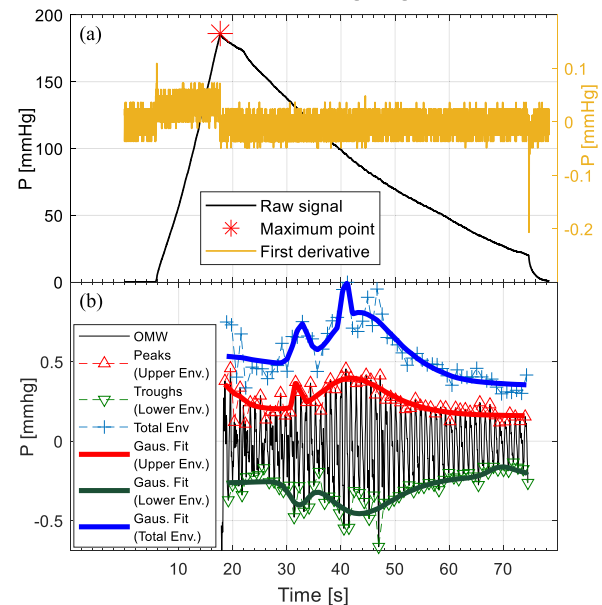


Fig. 2. Oscillometric waveform (OMW) measurement and modeling. It is observed that the shape of the oscillometric waveform envelope deviates from the Gaussian model for the diseased individual. As a result, it is expected that the model parameters including the center, amplitude, and spread parameters as well as the error statistics can be used to characterize PAD from healthy patterns.

TABLE II

LIST OF MODEL PARAMETERS AND ERROR STATISTICS EXTRACTED FROM OSCILLOMETRIC WAVEFORM ENVELOPES

Parameters/Envelope	Amplitude	Center	Spread
Upper	a_i^U	μ_i^U	σ_i^U
Lower	a_i^L	μ_i^L	σ_i^L
Total	a_i^T	μ_i^T	σ_i^T
Goodness-of-Fit/Envelope	SSE	R-Squared	Adjusted R-Squared
Upper	SSE^U	RSQ^U	$aRSQ^U$
Lower	SSE^L	RSQ^L	$aRSQ^L$
Total	SSE^T	RSQ^T	$aRSQ^T$

Subscript i correspond to the i th Gaussian function parameters.

i.e., when the applied cuff pressure is equal to mean arterial pressure [28]. Variations in pulse amplitude over the cuff deflation period are the result of nonlinear geometric collapse and elastic distention of the artery at different transmural pressures [24], and therefore their pattern can be used to characterize the arterial health and disease.

Envelope Detection: To characterize the pulse amplitude variations over the cuff deflation period, the upper, lower, and the total envelopes of the OMW were obtained by detecting the peaks and troughs of the oscillometric pulses (Fig. 2(b)). An initial estimate of the peak-to-peak interval was obtained by calculating the power spectral density of the OMW. The main peak of OMW power spectral density in frequency range of 0.6 Hz to 1.5 Hz was taken as an estimate of the inverse peak-to-peak interval. The OMW peaks and troughs were then defined as the local maximums and minimums of the OMW that are at least half of the estimated peak-to-peak interval apart. The patterns of the extracted envelopes at different cuff pressures were further studied to characterize the arterial system characteristics and to detect PAD.

C. Recognition of PAD Patterns

Mathematical Modeling: It has been found through several studies that the OMW envelope resembles a Gaussian shape and the model parameterization varies in different cardiovascular conditions [29], [30]. Based on this observation, the extracted OMW envelopes were mathematically modeled as a sum of N_g Gaussian functions (Fig. 2(b)). The model possesses $3 \times N_g$ parameters including center μ , spread σ , and amplitude a as listed in Table II. To find the optimum model parameters, a least-squares curve fitting was performed between the mathematical model f and the extracted envelopes $\tilde{f}$, as follows:

$$\{\hat{a}_i, \hat{\mu}_i, \hat{\sigma}_i\}_{i=1:N_g} = \arg \min_{\{a_i, \mu_i, \sigma_i\}_{i=1:N_g}} \sum_{cp} [\tilde{f}(cp) - f(cp)]^2 \quad (1)$$

where f is defined as follows:

$$f(cp) = \sum_{i=1}^{N_g} a_i e^{-\frac{(cp - \mu_i)^2}{2\sigma_i^2}} \quad (2)$$

and cp represents the cuff pressure. The minimization process shown in (1) was performed using trust-region algorithm [31].

To find the optimum number of Gaussian functions N_g , the degree of freedom adjusted R-square statistic was calculated. R-square statistic measures the successfulness of the fit in explaining the variation of data. It is defined as the square of the correlation between the measured envelopes and the fit as follows:

$$RSQ = 1 - \frac{\sum_{cp} w_{cp} [\tilde{f}(cp) - \hat{f}(cp)]^2}{\sum_{cp} w_{cp} [\tilde{f}(cp) - f_m]^2} = 1 - SSE/SST \quad (3)$$

where f_m represents the mean of the measured envelopes and $\hat{f}$ represents the fitted model. The denominator term in the second term of R-square formula, is called the total sum of squares (SST). R-square varies from 0 to 1, with a value closer to 1 indicating that a greater proportion of variance is captured by the model. The adjusted R-square is a modified version of the R-square statistic that is adjusted for the number of parameters in the model (K) relative to the number of data points (N) defined as follows:

$$aRSQ = 1 - \frac{SSE/(N - K - 1)}{SST/(N - 1)} \quad (4)$$

If new sets of parameters are introduced into the model one at a time, the level at which the adjusted R-square reaches a maximum, and decreases afterward, would be the model with the best fit without unnecessary terms. Here, the average R-square was calculated on a validation set consisting of data collected from two patients with PAD and two healthy subjects (see Section II-D). The optimal model was selected as the one with the highest number of Gaussian functions at which the adjusted R-square did not increase more than 1% compared to the model with one less Gaussian function. Sum of four Gaussian functions ($N_g = 4$) was found to achieve the optimal results with averaged adjusted R-square of 0.96 on the validation set.

The derived model parameters represent the best Gaussian fit to the measured waveform, and therefore were used to concisely characterize the arterial system behavior at different cuff pressures. In addition to the model parameters, the deviation of the measured envelopes from the model was also characterized with three goodness-of-fit statistics. These statistics represented how well the model can explain the arterial behavior at different cuff pressure. They included R-square (defined in (3)), adjusted R-square (defined in (4)), and sum of squared estimate of errors (SSE) defined as follow:

$$SSE = \sum_{cp} w_{cp} [\tilde{f}(cp) - \hat{f}(cp)]^2 \quad (5)$$

where $\hat{f}(cp)$ represents the optimum model derived according to (1) and w_{cp} is the weighting applied to each data point (here set to 1). A value closer to 0 indicates that the model better predicts the arterial behavior, and that the fit will be more useful for prediction.

Feature Engineering: The model parameters of the three extracted envelopes (36 parameters per each measurement) and

their error statistics (9 parameters) were treated as features characterizing the arterial system behavior in different transmural pressures. The features were ranked based on the amplitude of their corresponding Gaussian function to form a feature vector of dimension 45.

To help the classifier train faster and reduce the chances of getting stuck in local minima, a z-transformation was applied to all features by removing the mean and scaling based on the standard deviation. To further study the efficacy of each parameter in characterizing the arterial system behavior for the detection of PAD, GA was used as a preprocessing step. GA is a search heuristic method that works by selecting the best subset of features through a random selection process [32]. Inspired by the theory of natural evolution, each subset of features is represented by a binary chromosome and new feature generations are made based on crossover and mutation operators. This algorithm resembles the process of natural selection where the fittest features are selected for reproduction in order to produce offspring of the next feature generation.

A multilayer perceptron was trained to predict the class labels (here normal and PAD patterns) using the selected features. Classifier's mean square error (MSE) was used as a measure of its performance. A multi-objective function defined as the combination of classifier mean square error and features ratio, was used to determine the fitness of selected feature subset in each generation, as follows [33]:

$$O(H_i) = \gamma \cdot mse(H_i) + (1 - \gamma) \cdot \frac{d(H_i)}{d} \quad (6)$$

where $mse(H_i)$ is the classifier error, $d(H_i)$ is the number of selected features in chromosome H_i , d is total number of features, and $\gamma \in [0, 1]$ (here set to 0.8) is a coefficient that makes tradeoff between the two terms of objective function.

The first term in the objective function is related to the classifier error and the second term is the ratio of the number of selected features to the total number of features. This multi-objective function ensures that the least number of features with best pattern predictivity are selected. The objective function in (6) was iteratively minimized over different generations created with usual crossover and mutation operators to find the best feature set. The algorithm was set to stop when the average relative change in the objective function value over 50 generations was less than $1e-6$.

To further reduce the dimensionality of data, principal component analysis (PCA) was applied on the GA-extracted features. PCA transforms data into an orthogonal subspace, where the axes of this new coordinate system are defined in the direction of maximum variance of the original data. Those transformed features that explained $p\%$ (here 99%) of data variance were retained and the rest were discarded.

PAD Classifier: An ensemble classifier was designed to identify PAD patterns from normal based on the GA extracted features. Given our data is imbalanced, meaning PAD class in the training data has fewer samples than the normal class, the random undersampling boosting (RUSBoost) technique was employed to generate an ensemble model [34]. The base learner was set to a multilayer perceptron (MLP) NN. MLP does not

require an explicit mathematical model between the extracted features and their class labels and therefore is suitable for characterizing the arterial system behavior that defy modeling due to its nonlinear and complex nature. It has been shown that a two-layer feed-forward NN can approximate any function with finite number discontinuities if enough number of neurons exists in the hidden layer [35]. An MLP with two layers was selected as the base learner to distinguish PAD patterns from normal. The hyperbolic tangent sigmoid activation function was used in the input and hidden layers to provide enough network flexibility to learn the complex input-output patterns, and a softmax activation function was used in the output layer to assign a class probability to each input feature set. In case of multiple measurements from the same individual, the output of softmax layers were average to form the class probabilities (see Section III-D). The number of neurons in the input layer was equal to the number of input features, the number of neurons in the hidden layer were set to five based on the best results obtained on our validation set, and the number of output layer neurons was one. The network was first trained using the extracted features as input, and class label 1 (representing PAD) or 0 (representing normal) as the corresponding target. Once the network extracted information on the association between the input (extracted features) and target (PAD vs normal), it was used to estimate find PAD pattern from any new input pattern that has not been provided during training. The training was performed in batch style using Levenberg-Marquardt algorithm with backpropagation technique. The network performance function was set to cross entropy between its output and the desired target. The maximum number of training epochs was set to 1000 and training process was set to stop if the change in performance function value over six generations was less than $1e-6$. The number of base learners was set to 20 and the final class label for each input pattern was obtained using an equal-weight majority vote by all base learners.

The hyper-parameters of the proposed PAD detection system included GA coefficient γ , PCA cutoff parameter p , and number of hidden neurons NH , number of NN training epochs NT , and the number of base learners NB . The Random hyper-parameter optimization search [36] was employed to sample each hyper-parameter from a predefined range and to find the optimum configuration with the smallest validation error. The optimum values for γ , p , NH , NT , and NB were found to be 0.8, 0.99, and 5, 1000, and 20, respectively, based on validation data (see Section II-D).

D. Data Analysis

To evaluate the performance of the PAD detection algorithm, our collected data were divided into five groups: data collected from upper limbs of healthy individuals, lower limbs of healthy individuals, diseased lower limbs of individuals diagnosed with PAD, normal lower limbs of individuals diagnosed with PAD, and normal upper limbs of individuals diagnosed with PAD. The performance of the algorithms was evaluated on detecting PAD from diseased lower limbs as well as normal upper limbs of individuals diagnosed with PAD. The invasive angiography

technique was used as the gold standard to detect the diseased limbs [6].

Data from four randomly selected subjects (two patients with PAD and two healthy subjects) were used to form a validation set used to find the model hyperparameters. The performance of the proposed method was then evaluated by leaving two subjects out for test (one healthy and one with PAD) and the procedure was repeated so that all the combinations of two subjects were once left out for test. The estimated class probabilities for each test subject were then averaged. The results obtained on the test set was reported.

True positive (TP), false positive (FP), false negative (FN), and true negative (TN) rates were calculated according to the true class labels. The performance of the algorithm was evaluated in terms of overall accuracy (ACC), overall error (E), sensitivity or true positive rate (TPR), specificity or true negative rate (TNR), precision or positive predictive value (PPV), and negative predictive value (NPV).

The performance of the proposed method was compared with the conventional ABI technique. The ABI value was determined by taking the systolic pressure at each ankle, divided by the average brachial arterial systolic pressures. The blood pressure reading was performed simultaneously with the oscillometric recording using the Omron blood pressure monitor. The lower limbs of individuals with an ankle-brachial index of below a threshold were identified to be affected with PAD [15]. The optimal threshold value was found by leaving two subjects out for test (one healthy and one with PAD), and the optimum threshold was found on the rest of the subjects by changing the threshold value from 0.8 to 1.2 (steps of 0.01) and selecting the threshold value that led to the best classification accuracy. The procedure was repeated so that all the combinations of two subjects were once left out for test. The average optimal ABI was found to be 0.98 in our dataset.

To regress out the effect of age, a linear multiple regression analysis was performed where age and PAD class were used as independent variables and each of the 12 most important features were once selected as the dependent variable. The effect of age was then regressed out from the features by removing age component from the model. To analyze feature importance, each feature was once randomly shuffled, and the model error was computed. This process was repeated 10 times for each feature and the average model accuracy was calculated. The features were then ranked in importance according to the amount of decrease in PAD detection accuracy [37]. The kernel densities for the features were plotted after removing the effect of age and the feature statistics were reported.

III. EXPERIMENTAL RESULTS

A total of 232 oscillometric waveforms from upper and lower limbs were analyzed to detect PAD. Out of the analyzed waveforms, 40 belonged to diseased lower limbs of individuals diagnosed with PAD, 8 to normal lower limbs of individuals diagnosed with PAD, 48 to normal upper limbs of patients diagnosed with PAD, 68 to normal lower limbs of healthy individuals, and 68 to normal upper limbs of healthy individuals.

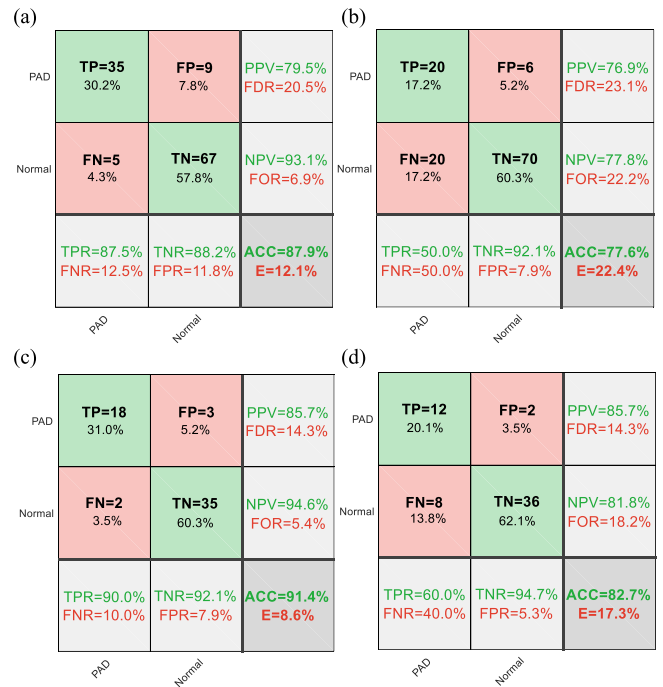


Fig. 3. Confusion matrices of (a) proposed method and (b) conventional ankle-brachial index (ABI) test using a single oscillometric measurement for the identification of PAD from diseased lower limbs. The same confusion matrices when combining two repetitive measurements are shown in (c) and (d), respectively.

Our developed algorithm was applied to identify diseased lower limbs of patients with PAD (40 patterns) from normal lower limbs of healthy individuals as well as PAD patients (76 patterns). Out of 40 PAD patterns, the proposed algorithm correctly detected 35 patterns as PAD (TP) and misclassified only 5 patterns as healthy (FN). Out of the 76 normal patterns, 67 were correctly classified as healthy (TN) and only 9 were misclassified as PAD (FP) (see confusion matrix in Fig. 3(a)).

The proposed algorithm achieved an overall accuracy of 87.9% and overall error of 12.1%, which shows its overall reliable performance in distinguishing between PAD and normal oscillometric patterns. The proposed algorithm achieved a high sensitivity of 87.5% showing that most of PAD patterns are correctly detected. It also achieved a high specificity of 88.2% showing that most of normal patterns are correctly identified as such. The achieved NPV of the proposed algorithm was 93.1% showing that most of the detected normal patterns are truly normal. The achieved precision was 79.5% showing that a large fraction of detected PAD patterns is truly PAD. The algorithm's F-score, which determines the harmonic mean of precision and sensitivity, was 83.3% which shows the algorithm's good performance in maximizing both precision and sensitivity at the same time.

A. Ankle-Brachial Index (ABI) Test

In comparison to our proposed method, ABI test achieved an overall accuracy of 77.6%, overall error of 22.4%, sensitivity of 50.0%, specificity of 92.1%, precision of 76.9%, NPV of 77.8%, and F-score of 60.6% (see Fig. 3(b)). It is observed that

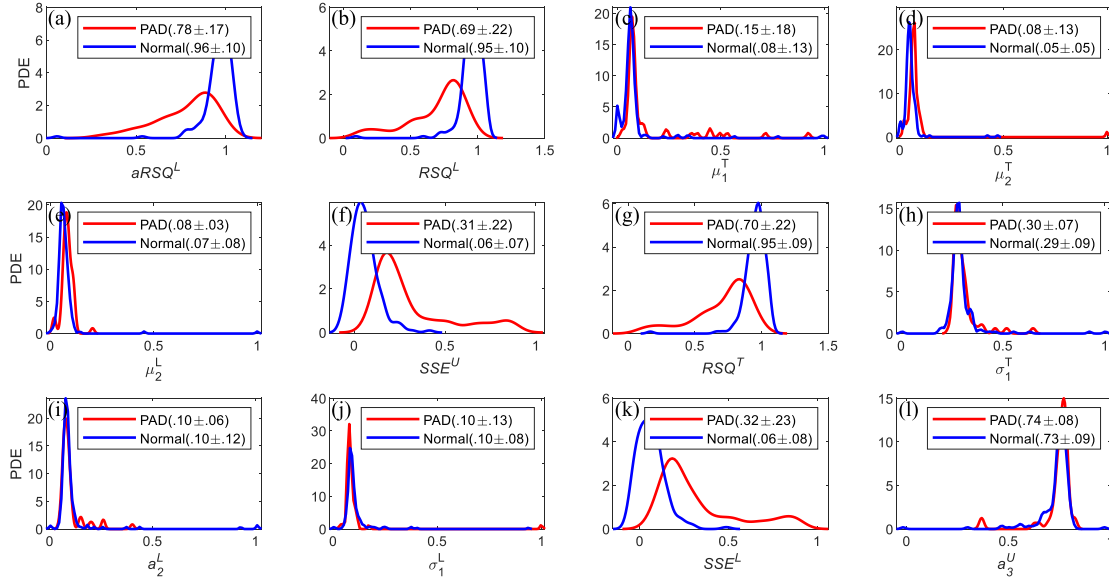


Fig. 4. Kernel density plots of the most important features for the detection of PAD. The features are ranked based on their importance from a to l.

the ABI test was not as successful as our proposed approach in detecting patients with PAD (37.5% lower sensitivity). The accuracy, precision, and NPV of the ABI test were also 10.3%, 2.6%, and 15.3% lower compared to the proposed method. However, the ABI test could specify normal subjects slightly better than our proposed method (3.9% higher specificity).

B. Effect of Multiple Measurements

The performance of the proposed method was also verified when information from two repetitive measurements (taken with 10 mins delay) from an individual were combined for the detection of PAD. The same PAD classifier was applied to find PAD probability for each measurement and the estimated probabilities were averaged to find the final PAD probability. The confusion matrix for the proposed PAD detection method for the case of combining two repetitive measurements is shown in Fig. 3(c). For the conventional ABI test, blood pressures were average over multiple measurements and the averaged values were used to detect PAD. The confusion matrix of the ABI test after combining multiple measurements is shown in Fig. 3(d).

It was observed that the accuracy of our proposed method in the detection of PAD was improved to 91.4% (3.5% improvement). The sensitivity, specificity, precision, and NPV were also improved by 2.5%, 3.9%, 6.2%, and 1.5%, respectively. The accuracy, sensitivity, specificity, precision, and NPV of the ABI test were improved by 5.1%, 10%, 2.6%, 8.8%, and 4%, respectively. These results show the potential to improve PAD detection when fusing information from multiple measurements.

C. Most Effective Features

The most effective features selected with GA for the detection of PAD from lower limbs according to their importance were $aRSQ^L$, RSQ^L , μ_1^T , μ_2^T , μ_2^L , SSE^U , RSQ^T , σ_1^T , a_2^L , σ_1^L , SSE^L , and a_3^U , respectively (see Table II for the full list of features). Among the 12 extracted features, five were error

statistics (ranked 1st, 2nd, 6th, 7th, 11th) among which two had the most importance in the detection of PAD showing that the deviation of oscillometric waveform envelopes from a Gaussian shape can represent arterial disease (see Fig. 2). Three center parameters were also selected as important features (ranked 3rd, 4th, 5th) showing the location of oscillometric waveform envelope peaks could vary from healthy to diseased patterns (see Fig. 2). Two spread parameters (ranked 8th, 10th) and two amplitude parameters (ranked 9th, 12th) were also among important features for the detection of PAD. As shown in Fig. 2, the Gaussian fit to the diseased pattern produces a less accurate fit (leading to higher error statistics), exhibits several separated peaks (leading to different center parameters), and has heavier tail (leading to different spread and amplitude parameters).

Kernel density plots for the selected 12 features are shown in Fig. 4, where mean and standard deviations for healthy individuals and patients with PAD are listed on each plot. The plots are sorted according to feature importance, where features with higher importance are plotted first. It is observed that error statistics have the highest separation between healthy and PAD patterns followed by center parameters, spread parameters, and amplitudes.

IV. DISCUSSION

Peripheral arterial disease (PAD) is an important risk factor for cardiovascular disease that if not detected and treated early can lead to serious cardiovascular complications such as heart attack or stroke. Most of existing PAD diagnosis tools are invasive, cumbersome, time-consuming, and require expensive equipment and highly trained technicians.

Our proposed method was found to achieve higher accuracy (8.7%), sensitivity (30.0%), NPV (12.8%), and F-score (17.2%) compared to the conventional ankle-brachial index (ABI) test when comparing their best performances on multiple measurements. However, the ABI test achieved higher specificity

(2.6%). The precision of the proposed method and the ABI test were equal. The reason for low sensitivity of ABI test can be associated to the calcified arteries in patients with diabetes that make the arteries uncompressible leading to high values of ABI [38]. The overall accuracy of the proposed method was 8.6% higher on multiple measurements and 10.3% higher on single measurements when compared to the ABI test.

The accuracy of the conventional ABI approach partly relies on the accuracy of blood pressure estimates obtained on lower and upper limbs. However, the estimation of blood pressure using automated oscillometric devices is challenging as such monitors are mainly designed for individuals without PAD. Any variation in the oscillometric waveform shape from its normal pattern may lead to inaccurate estimates of the blood pressure. However, unlike the ABI method, our proposed method does not rely on blood pressure estimates, but it takes advantage of the oscillometric waveform pattern variations to detect PAD. Mathematical modeling, feature extraction, and machine learning was used to detect those abnormal oscillometric patterns corresponding to PAD. Our results reported in Fig. 3 show the high accuracy of our proposed method compared to the oscillometric-based ABI technique.

In addition to the detection of PAD using data collected from diseased lower limbs, our method was also evaluated on the detection of PAD from normal upper limbs (arms) of individuals affected with PAD. Given that PAD is usually a progressive disease generalizing to the whole cardiovascular system [39], the idea was to verify if data collected from normal limbs of patients with PAD can reveal the disease. The achieved accuracy, sensitivity, and specificity were 74.1%, 75%, and 74.5%, respectively. The proposed technique may be improved by finding a transfer function between the measurements obtained from the upper and lower limbs. Given that the arterial waveform on upper limbs is affected by the wave reflection arriving from lower limbs, the changes in the lower limb arterial system may be reflected in the upper limb measurements [40]. The proposed approach may also be extended toward detecting arterial disease in central arteries such as carotid, aorta, and iliac arteries. For example, transmission line models can be derived between central and peripheral pulse waveforms for the detection of cardiovascular disease [41]–[43]. Adding auxiliary sensors such as ECG can further improve such models [44], [45]. Another possible approach is the use of machine learning and deep learning algorithms to find the implicit relationship between the peripheral and central waveforms [17].

Given that performing multiple measurements may not be convenient to the user and are time consuming, here, our main focus was on using a single oscillometric reading for the diagnosis of PAD. We, however, performed our classification task, by considering all the four measurements together. The results were slightly improved by about 1%.

Our proposed technique was based on modeling the recorded oscillometric waveforms, extracting features, and designing classifiers to detect PAD. In addition to the proposed method and ABI test, the performance of a support vector machine (SVM) classifier and a decision tree (DT) classifier were also investigated in the detection of PAD from the 12 extracted

features. SVM and DT achieved accuracies of 87.9% and 86.2% in the detection of PAD from multiple measurements obtained from lower limbs which were 3.5% and 5.2% lower than the proposed method. The performance of the proposed method may be improved by employing deep learning algorithms that extract representative features from the raw data without the need for any feature engineering. Collecting a large dataset of oscillometric recording from healthy individuals as well as patients with PAD is necessary to develop such a deep learning PAD detection method.

Sum of four Gaussian functions were found to best fit the oscillometric waveform upper, lower, and total envelopes. The amplitude, center, and spread parameters of these Gaussian functions along with three error statistics were used as features best characterizing arterial system behavior. Among the extracted features, GA selected 12 features as the most informative for the detection of PAD from diseased lower limbs. Out of the extracted features, the error statistics were mostly selected showing that any model deviation from the Gaussian shape (that is explained through the error statistics) could be representative of a diseased arterial system behavior. The other mostly selected features were center parameters followed by spread parameters and amplitudes showing that the location, spread, and the amplitude of Gaussian functions can also characterize the healthy versus diseased arterial system behavior. For normal limbs, the Gaussian functions are usually centered around a cuff pressure equal to the mean arterial pressure, i.e., when the artery is unloaded [29], while for diseased limbs the Gaussian functions tend to spread.

An advantage of the proposed method is that it does not require any special or additional transducers or sensors but those already being used in oscillometric blood pressure monitors. Thus, PAD detection can be performed by an untrained observer or by the user along with routine blood pressure oscillometric readings. In contrast, most of the existing PAD detection techniques are invasive [6], [7], not fully automated [10], [11], or require expensive sensors and transducers [13]. It should be noted that while the conventional ABI test may be performed by automatic oscillometric blood pressure monitors [15], it relies on four measurements performed on arms and ankles.

For early detection of PAD, affordable, noninvasive, and easy-to-use devices are required that can provide relatively immediate information about peripheral arterial functionality. Our proposed approach can be easily integrated into the automated oscillometric blood pressure devices and therefore can be adopted for in-home cardiovascular health monitoring due to its relatively low-cost, minimum training requirement, and the ability to detect PAD by performing a simple oscillometric measurement. Our proposed approach can be used as a reliable initial diagnosis or screening tool replacing ABI test before performing a final diagnostic test such as angiography. A reliable initial PAD diagnosis method is one with high sensitivity where the number of FNs are as low as possible. The individuals initially detected with PAD can be referred to a specialist for further evaluation and diagnosis. Our proposed method misclassified only 2 out of 20 PAD patterns as normal and achieved a high sensitivity of 90.0% making it a reliable and easy-to-perform initial test for the detection of PAD.

In addition to the detection of PAD from single oscillometric measurements, the effect of multiple measurements was also studied by combining information from two repetitive measurements. Detection accuracies of our proposed method and the ABI test were improved by 3.5% and 5.1%, respectively, when using data from multiple measurements. This shows the potential to achieve higher reliability and accuracy in the detection of PAD when fusing information obtained from several measurements.

A correlation analysis was performed between the classifier probability score and the subjects' age. For PAD patients, a positive correlation coefficient of 0.40 was observed and the correlation was statistically significant ($p < 0.01$). This implies that according to our classifier, older individuals have higher probability of having PAD. For healthy individuals, a negative correlation coefficient of -0.02 was observed which was not statistically significant ($p = 0.83$). A correlation analysis was also performed between the ABI and the classifier score for all individuals. The correlation coefficient was 0.59 which was statistically significant ($p < 0.001$), meaning that there was a close agreement between our classifier output probability score and the ABI.

In this study, oscillometric measurements were performed with an automatic cuff (Omron 7321 M6) that was set to deflate from a predetermined suprasystolic pressure to a subdiastolic pressure at a fixed rate of 2 mmHg/s. As soon as the cuff pressure reached the predetermined subdiastolic pressure, it was promptly deflated. The recording time from the beginning of cuff deflation to a complete deflation was 30 ± 6 s and the recording time from the beginning of inflation to the complete deflation was 50 ± 9 s over all the measurements. For user comfort, the cuff deflation rate may be increased, or the recording may be performed for a shorter time interval. However, further study is required to evaluate the performance of the proposed method in these settings.

Arterial disease affects the arterial wall dynamics and its behavior at different transmural pressures [46]. As the arterial disease progresses the arteries lose their elastic properties and become stiffen. It has been shown that the change in the amplitude of noninvasively measured arterial pulses at different cuff pressures can be used to characterize the arterial elastic properties [24], [26]. Therefore, we relied on the analysis of the oscillometric waveform envelope for the detection of PAD. It should however be noted that the oscillometric waveform envelopes may vary in shape across different individuals independent of PAD. Different factors such as arm, cuff, and artery characteristics as well as blood pressure level can lead to such variations. Here, we extracted the most relevant features of the oscillometric waveform envelopes that lead to the best classification results (using a validation set). We then trained a classifier to find the implicit relationship between the target class (PAD vs healthy) and the extracted features. This approach was tuned using a validation set, trained on a training set, and tested on data not seen during model tuning and training. The results obtained on the test data were then reported. Therefore, it is expected that the proposed approach is generalizable to new unseen data. Our model may be further improved if a larger number of data collected from individuals

in different age groups and health conditions as well as different recording settings is used for training. Using deep learning algorithms that automatically perform the feature extraction and classification without the need for expert designed features can also improve the reliability and generalizability of our approach.

The comparison between healthy and PAD groups may have been confounded by the large differences in age between the healthy subjects and patients with PAD ($p < 0.05$). Future work should be directed toward evaluation of the proposed approach in distinguishing PAD from healthy patterns on larger groups of healthy individuals and patients affected with PAD with similar age statistics.

V. CONCLUSION

We presented a novel method for the accurate detection of PAD using an oscillometric cuff. The method was based on characterizing the arterial system normal versus diseased behavior by applying an external time-varying pressure using a cuff and measuring the superposition of arterial and cuff pressures. The arterial system behavior at different cuff pressures was mathematically modeled and the model parameters were incorporated to develop a machine learning approach for the automatic detection of PAD.

The performance of our proposed method was validated for the detection of PAD from diseased lower limbs of patients affected with PAD and was found to outperform the conventional ABI-based method. However, the effect of age as a confounding factor was not considered. The proposed method shows promise toward the development of unexpensive, easy-to-use, and reliable systems for the early detection of PAD.

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